Lab 3: Architecture and their costs

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Counting convention:

I used a batch size of 1. One multiply-accumulate operation, or MAC, counts as two FLOPs: one multiplication and one addition.

I counted matrix multiplications and convolutions. Biases are included in the parameter counts, but bias additions and other operations, such as ReLU, pooling, normalization and softmax, are excluded from the FLOP calculations.

MLP: first linear layer

The first linear layer has 784 input features and 256 output features.

Number of parameters:

$$P = (784 \times 256) + 256$$

$$P = 200,704 + 256 = 200,960$$

Number of MACs:

$$MACs = 784 \times 256 = 200,704$$

Number of FLOPs:

$$FLOPs = 2 \times 784 \times 256$$

$$FLOPs = 401,408$$

This is the cost of the first linear layer for one image.

CNN: first convolutional layer

The input has one channel and a size of 28×28. The layer has 16 filters of size 3×3, with stride 1 and padding 1.

Output height and width:

$$H_{out} = W_{out} = \left\lfloor \frac{28 + 2 \times 1 - 3}{1} \right\rfloor + 1 = 28$$

The output shape is:

$$16 \times 28 \times 28$$

Number of parameters:

$$P = (3 \times 3 \times 1 \times 16) + 16$$

$$P = 144 + 16 = 160$$

Number of MACs:

$$MACs = 3 \times 3 \times 1 \times 16 \times 28 \times 28 \qquad MACs = 112,896$$

Number of FLOPs:

$$FLOPs = 2 \times 3 \times 3 \times 1 \times 16 \times 28 \times 28 \qquad FLOPs = 225,792$$

The same filter weights are used at different positions in the image. This keeps the parameter count small, but each position still requires calculations.

Lab 3: Hand calculations

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Attention: one self-attention module

The image is divided into patches of size 7×7

Number of patches: $N = \frac{28}{7} \times \frac{28}{7} = 4 \times 4 = 16$

The embedding dimension is $d = 64$, with four attention heads.

Dimension per head: $d_{head} = \frac{64}{4} = 16$

Number of parameters, including the QKV projections and the output projection:

$P = 3d^2 + 3d + d^2 + d$ $\quad P = 3 \times 64^2 + 3 \times 64 + 64^2 + 64$ $\quad P = 12,288 + 192 + 4,096 + 64$

$P = 16,640$

Q, K and V projections

$FLOPs_{QKV} = 2 \times N \times d \times 3d = 2 \times 16 \times 64 \times 192 = 393,216$

Attention scores: QK^T

$FLOPs_{scores} = 2 \times N^2 \times d = 2 \times 16^2 \times 64 = 32,768$

Attention weights multiplied by V

$FLOPs_{AV} = 2 \times N^2 \times d = 2 \times 16^2 \times 64 = 32,768$

output projection

$FLOPs_{output} = 2 \times N \times d^2 = 2 \times 16 \times 64^2 = 131,072$

Total self-attention cost:

$FLOPs = 393,216 + 32,768 + 32,768 + 131,072$ $\quad FLOPs = 589,824$

$MACs = \frac{589,824}{2} = 294,912$

The four heads are already included because the formulas use the total embedding dimension of 64. This calculation covers one self-attention module. It does not include the encoder's separate feed-forward layers or normalization.

Comparison with Thop

Layer	Hand-calculated FLOPs	THOP MACs	THOP FLOPs
First linear layer	401,408	200,704	401,408
First convolution	225,792	112,896	225,792
Self_attention	589,824	294,912	589,824

I multiplied the THOP MAC counts by two to compare them with my FLOP calculations. For the linear and convolutional layers, the counts match using THOP's default rules. Default THOP does not fully count MultiheadAttention, so the attention result uses a custom rule covering the QKV projections, both attention matrix products and the output projection. The custom rule uses the same counting convention as these calculations. The matching results show that the calculations are consistent, but they are operation estimates, not hardware measurements.